

Customer Sphere Design

HawkinsTwin Customer 360 — an indexed, ACDOCA-aligned sphere of vectored targets

PREPARED FOR

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Hawkins Truck Mart's customer records live in a dealer management system, a call platform, and the systems Craig Allen runs day to day. This design resolves all of them onto one customer, in one combined schema, with one key. It is built as an index repository inside the wider enterprise reference model, extending the existing ACDOCA standard at the edges rather than replacing it, with every face of the customer carried as a vector against a declared target. The description of the model is published and portable. The data itself stays inside Peterbilt Atlantic's own Azure tenant and never leaves it. That split is the whole design, and it is what puts the dealership — not a vendor — in the reference-model position for its industry.

Status — Draft for review. No dealership data has moved.

Agreement holder Tim Hawkins · **20 Group access** Luke Weatherbie · **Systems register** Craig Allen

Source of record [customer-sphere/CUSTOMER-SPHERE-DESIGN.md](#)

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Programme: EVEglyphDesign — Peterbilt Atlantic Digital Twin **Direction record:** 2026-08-01 — Customer 360 and the reference-model position **Agreement holder:** Tim Hawkins · 20 **Group access:** Luke Weatherbie · **Systems register:** Craig Allen **Constraint:** the live CDK dealer management system is never written to. **Status:** draft. No dealership data has moved.

1. Why a sphere and not a pipeline

A pipeline has a direction. Data leaves a system, is transformed, and lands somewhere downstream where someone looks at it. That shape is adequate for reporting and inadequate for a customer, because a customer is not downstream of anything. The same fleet owner is simultaneously a sales record in CDK, a parts account, a service history, an open warranty claim, six unreturned voicemails, a finance file, and a name Craig's marketing systems know under a different spelling.

The customer sphere is the surface on which all of those faces resolve to one entity. The integrator's job, as it has been stated across this programme, is to hold economics, operations, and relationship in a single event horizon and let the sphere emerge from the triangulation. This document is that principle applied to one dealership's customers.

Practically: **one combined schema, one customer key, many faces.** No face is authoritative over the others. Each face carries its own source, its own timestamp, and its own confidence, and the sphere records disagreement rather than averaging it away.



2. The two-body split

This is the governing constraint of the design and every other decision is subordinate to it.

	Index repository	Business data store
What	Registry of what exists, field dictionary across every vocabulary, identity rules, declared targets, lineage, model definitions, DDL, validators, tests, change history — pointers, never payload	Actual customer, sales, service, parts, finance and call records
Where	GitHub — HawkinsTwin Customer 360 index repository	Postgres, inside Hawkins's own Azure tenant
Custody	EVEglyphDesign authors, Hawkins owns; readable by any party that needs to understand the model	Hawkins alone. EVEglyphDesign holds no copy
Contains a customer record?	Never	Yes, all of them
Portable?	Yes, deliberately — it is the reference model	Yes, but it moves only on Hawkins's instruction

The split answers the question every dealership principal eventually asks, which is not *what does it do* but *who ends up holding my data*. The answer here is that the description of the data is public and forkable, and the data itself never leaves the tenant Hawkins already pays for and already controls. A vendor cannot hold the dealership hostage with a schema it refuses to document, and cannot hold it hostage with data it refuses to hand back, because it never had either.

What the GitHub side actually is, and how it routes, is §3. A secondary consequence matters for the commercial case in §11: because the index is complete enough to rebuild the model from scratch, the model is separable from the data. That is what makes it sellable as a reference model without selling anyone's customers.



3. An index repository, not a data repository

The GitHub side of the split is more precisely described than "metadata". It is an **index repository**, and it behaves the way every index in the EVEglyphDesign portfolio behaves: it is the entry point you consult *before* you touch anything, and it holds pointers, not payload.

An index repository answers four questions and refuses to answer any others:

- **What exists.** Every source, object, and field in the customer sphere, registered once with an owner, an access route, and a confidence tag. If something is not in the index, it is not in the sphere — and if it exists but cannot be reached, the index says so rather than omitting it.
- **What it is called, in every vocabulary it has.** The CDK name, the call-platform name, the name Craig's system uses, the canonical name in the model, and the SAP-side term it aligns to. One field, several names, one entry.
- **Where the real thing lives.** A pointer into the Azure Postgres — schema, table, column — never the value itself.
- **What governs it.** Retention, personal-data classification, the gate that applies, and the decision record that put it there.

This matters operationally. Anyone — a developer at Hawkins, a PACCAR analyst, a future vendor, an agent doing work on the twin — starts at the index, is routed to the right definition and the right doctrine, and only then goes to the data. It is the same START-HERE discipline the rest of the portfolio runs on, applied to one dealership's customer estate. Navigation is a safety control: the index makes looking up the governing rule a hard gate rather than an optional search.

It also makes the repository forkable in a way a data repository never is. An index has no custody risk. It can be published, read, audited, cited, and copied by another dealership without a single Hawkins record moving, which is exactly what §11 depends on.



4. Where this sits in the enterprise model

The customer sphere is not a standalone Hawkins artefact. It is one instance inside the larger EVEglyphDesign enterprise model, and it inherits from that model rather than inventing beside it.

Layer	What it is	Role here
L0 — Portfolio index	The canonical entry point and routing layer across all EVEglyphDesign repositories	Routes any agent or reader to the governing doctrine before work begins
Enterprise reference model	EVE Datasphere Sovereign — SAP-parity schemas with ACDOCA as the accounting spine, held in customer custody	Supplies the vocabulary, the ledger spine, and the custody doctrine
Industry instance	The Peterbilt Atlantic digital twin	Applies the enterprise model to heavy-truck dealer operations
Customer sphere	This design — HawkinsTwin Customer 360	The customer-facing face of that instance, indexed and vectored
Business data	Postgres in Hawkins's Azure tenant	The only place a real record exists

The consequence is that nothing here is bespoke for its own sake. The customer key, the lineage discipline, the source-width rule, the widening register, the marts — all of them are portfolio patterns being instantiated, which is why a component lifted out of Hawkins lands cleanly in the next dealership instead of needing a rewrite.

Mirror non-proliferation applies. Source vocabulary wins. Structures are defined once and referenced. Extensions attach at the edges. The sphere does not get to invent a private dialect because it found one inconvenient.



5. An extension of ACDOCA, not a parallel standard

The accounting spine of the enterprise model is **ACDOCA**, the SAP universal journal. Every financially meaningful event in the sphere aligns to it, and this is the single most important structural decision in the design after the two-body split.

ACDOCA is never modified. Not extended in place, not widened, not given custom columns. It stays recognisable to anyone who knows SAP, which is what makes the dealership's numbers defensible to PACCAR, to a 20 Group, and to an auditor. What the sphere needs beyond the journal attaches as **extension journals at the edges**, in the pattern the enterprise model already establishes:

Journal	Carries	In the customer sphere
ACDOCA	The universal journal — financially meaningful transactions	Sales, F&I, parts invoices, warranty settlement, service revenue
ACDOCI	Service and relationship interactions	Calls, voicemails, callbacks, contact attempts, appointment events
ACDOCX	ESG and per-transaction impact measures	Available where a dealership wants transaction-level impact reporting
MRTDOC	Non-tradeable earned standing	Available; not part of Phase 1 for Hawkins
MEMBR	Membership and affiliation	Fleet-to-driver affiliation, account hierarchy

The important pairing is **ACDOCA and ACDOCI side by side**. The journal records what the dealership *sold*; the interaction journal records what the dealership *did about* the customer. Neither is derivable from the other, and a customer view built on only one of them is the reason most dealer CRM projects produce numbers nobody trusts. Sharing a customer key and a time axis across both is what makes "this account bought nothing for nine months and we never returned four of its calls" a single, defensible sentence.

Where CDK carries a field wider than the SAP default, **the source width wins** and the widening is recorded in a declared widening register. Dealer data is never truncated to fit the model. Canonical naming coexists with legacy names in the index, so relabelling never destroys round-trip fidelity back to the source system.



6. Vectored targets — the geometry of the sphere

"Sphere" is not decoration. It is the working geometry, and it is what makes the model computable rather than merely descriptive.

The customer sits at the centre. Every fact known about that customer is a **vector** from the centre — and a vector, not a field, because each one carries four things at once:

- **A direction** — which face it belongs to: commercial, service, parts, finance, warranty, relationship, marketing.
- **A magnitude** — how much it says. Revenue, hours, unreturned calls, days since last contact, warranty exposure.
- **An origin** — the source system, the extraction, the timestamp, the confidence. A vector with no traceable origin is not admitted to the sphere.
- **A target** — the state the business is aiming that face at.

That last one is the point. **A vectored target is a face of the customer with a declared destination.** Not a metric on a dashboard someone may or may not look at — a direction of travel, with a current position, a target position, and therefore a gap that can be measured, ranked, and dispatched to a person.

The 20 Group guide figures are targets. PACCAR's expectations are targets. The two-hour voicemail return is a target. Absorption is a target. Each one is a vector on the sphere with a known current magnitude and a known intended one, and the difference between them is the entire decision surface — the worklists, the three standing cohorts, the variance columns already specified in the twin.

Three properties follow, and they are the reason this shape was chosen:

1. **Composition.** Vectors on the same face add. Nine unreturned calls across four contacts at one fleet compose into one account-level relationship vector, without averaging the individual events away.
2. **Resolution of disagreement.** When two sources disagree, the sphere holds both vectors with their origins and confidences intact and reports the divergence. It does not collapse them into an average that hides which system was wrong.
3. **Ranking is intrinsic.** The largest gap between current and target, weighted by account value, is the top of the worklist. Prioritisation falls out of the geometry instead of being a rule someone maintains by hand.

Targets are declared in the index repository, versioned there, and never hard-coded into a model. Changing what the dealership is aiming at is a committed decision with a date and an author on it, not a silent edit to a query.



7. Sources into the sphere

Three source families, one schema.

7a. CDK — customer and sales

The dealer management system is the spine. Customer master, sales transactions, deals and F&I, repair orders, parts invoices, warranty claims, and the general ledger tie-out. Acquisition follows the existing two-lane design in the [reconciliation contract](#): the restored-backup lane for completeness, the API and export lane for currency, both landing in the same Postgres and reconciled on demand rather than silently merged. The field-level model — 21 objects and 443 confidence-tagged fields — is the starting dictionary for the customer and sales portion of the combined schema, not a separate artefact.

7b. Call history — the relationship record

The TELUS call platform supplies inbound and outbound call events, direction, duration, extension, disposition and voicemail state. This is the only source in the sphere that records what the dealership *did about* a customer rather than what it *sold* one. Its three standing cohorts — voicemail not returned within two hours, repeat caller without callback, and never reached — are the sphere's first behavioural signals. Ingestion runs under a Peterbilt-owned service identity so custody and audit trail stay with the company. Where call direction or extension identity is missing, the record contributes to cohort counts but is never used to score an individual.

7c. Craig's register — the operating surface

The sixteen systems catalogued by Craig Allen are treated as first-class sources, not as context. Marketing and lead platforms, inventory and ePortal surfaces, scheduling, and whatever else the register holds each contribute a face to the customer. Craig's register is the authority on what exists; this design's obligation is only to give each entry a defined landing in the schema, an owner, an access path, and a confidence tag — or an explicit note that it is deferred.

Open item. The source inventory is not closed until Craig confirms the register is complete and current, and until each system has a stated access route — API, export, or manual. Systems without an access route are recorded as known-but-unreachable rather than quietly dropped.



8. Identity — how a customer becomes one customer

Identity resolution is the hardest part of the sphere and the part most likely to be got wrong quietly, so it is specified before any model is built.

- **A durable internal key.** Every resolved entity carries a `customer_sphere_id` minted inside the sphere. No source system's primary key is promoted to the master key, because every one of them is a key for its own purposes and will eventually be reused, merged, or retired by its vendor.
- **Source keys are retained, never overwritten.** The CDK customer number, the phone identifier, and each of Craig's system keys are all kept against the resolved entity. Losing a source key destroys the audit path back to the record it came from.
- **Deterministic before probabilistic.** Exact matches on CDK customer number, on a normalised phone number, and on VIN-to-owner links are applied first. Only what remains unmatched goes to fuzzy matching on name, address, and business name.
- **Match confidence is stored, not thresholded away.** Each link carries a score and the rule that produced it. Low-confidence links are held as *candidate* links and surface in a stewardship queue rather than being auto-merged.
- **Merges are reversible.** Every merge is an appended, timestamped, attributed event. An unmerge restores both sides. A destructive merge is a defect, not a design choice.
- **Person and organisation are distinct.** A fleet is not a driver. The sphere carries both, with an explicit affiliation edge between them, because dealership relationships are almost always one organisation to several people.
- **Vehicle is a third axis.** Customer, organisation, and vehicle form the three anchors; service and warranty history hang from the vehicle, not from the person who happened to book the appointment.



9. Schema shape

Five layers, each with a job, all in the same Postgres and all described in the metadata repository.

1. **Landing.** Source-native. Original field names, original widths, nothing coerced. Where a source column is wider than the SAP-shaped default, the source width wins and the widening is recorded explicitly — dealer data is never truncated to fit a model.
2. **Conformed.** Types normalised, phone and address standardised, currency and dates unified, source keys preserved. Still one row per source record.
3. **Identity.** The resolution layer of §8 — entity tables, link tables, candidate queue, merge event log.
4. **Sphere.** The combined customer model. One customer, all faces: commercial history, service history, parts, warranty, finance, contact and call behaviour, marketing presence, and the cohort flags derived from them.
5. **Marts.** Analytical structures for the 20 Group performance set, service and parts performance, and the decision surfaces. These reuse the SAP-shaped analytical model and the [data mart library](#) rather than inventing a Hawkins-only structure — reuse is what lets a component be lifted to another dealership later without a rewrite.

Every field at every layer carries source, extraction timestamp, and confidence tag. A number that reaches a decision surface can always be walked back to the record and the extraction that produced it.



10. Governance and the gates

- **Nothing moves before the agreement.** Tim Hawkins's executed agreement gates any real dealership data. Until then the sphere runs on synthetic and mock composite data.
- **Peer comparison waits on access.** 20 Group peer figures populate only once Luke has access; blank variance fields stay blank rather than being guessed.
- **The live DMS is never written to.** Unchanged, in every phase.
- **Service identities, not personal credentials.** Every source connection is a company-owned identity with an attributable audit trail.
- **Tenant identity is checked centrally.** Every pull resolves the authenticated account and aborts on mismatch, in the shared token path, so no individual workflow can skip it.
- **Personal data stays in the tenant.** Names, phone numbers, addresses, account numbers and VINs exist only in Azure Postgres. The metadata repository holds field definitions and synthetic examples. Any commit that would place a real record in GitHub is a durability and privacy defect and is treated as one.
- **Retention and erasure are schema features.** Because Canadian privacy obligations apply to a New Brunswick dealership, the sphere carries retention policy per source and an erasure path that resolves through the identity layer, so a deletion request can be executed once and propagate rather than being chased across sixteen systems by hand.



11. Why this makes Hawkins the reference model

The commercial argument is not that the dealership gets better reports. It is structural.

A dealership that owns a fully described customer model — schema public, data private, lineage complete, models running in its own tenant — is in a position no vendor-hosted dealership can occupy. Its numbers are defensible to PACCAR and to a 20 Group because the derivation is inspectable. Its data is portable because the description of it is complete. Its components can be scaled and sold to peers without the underlying business data ever being part of the transaction.

That is the reference-model position: the industry ends up measuring itself against a shape that Hawkins owns, and Hawkins carries none of the custody risk that would normally be the price of being the example. Components scale commercially; the reference model stays with the dealership.



12. Open questions

1. Is the sixteen-system register complete and current, and what is the access route for each entry?
2. Which Azure subscription and region hosts the Postgres, and who administers it on the Hawkins side?
3. Does CDK entitlement permit the Data Export Tool over SFTP/PGP plus Dealer Data Exchange, which is the preferred route, or is a per-data-type third-party path required?
4. What is the retention policy per source, and who signs it off?
5. Which of Craig's systems hold customer contact data that duplicates CDK, and which is authoritative when they disagree?
6. Which extension journals are in scope for Phase 1? ACDOCA and ACDOCI are assumed; ACDOCX, MRTDOC and MEMBR are available but unscoped until Tim says whether impact reporting and account hierarchy matter to him now or later.
7. Who declares and signs off the targets — the 20 Group guide figures, the callback window, the absorption goals — and how often are they re-versioned in the index?



Provenance and control

Document

EVEglyphDesign — Hawkins Twin Customer Sphere Design, 14 pages.

Source of record

[customer-sphere/CUSTOMER-SPHERE-DESIGN.md](#) in the [Hawkins Twin Platform repository](#). The repository is authoritative; this PDF is a controlled copy of it at the timestamp below.

Direction record

[2026-08-01 — Customer 360 and the reference-model position](#), the dated exchange this design was written from.

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Standing gates

No dealership data moves before Tim Hawkins's executed agreement. Peer comparison waits on Luke Weatherbie's 20 Group access. The live CDK dealer management system is never written to. Every surface runs on synthetic or mock data until those clear.

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